

Analucia Vera Maric

AUTOMATION ENGINEER

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Systems engineer with 2+ years in IoT project management, industrial automation, and research. Background in Industrial Electronic Engineering; currently completing MSc Robotics & Automation. Skilled in IEC 61131-3 PLC programming, IoT protocols (OPC UA, MQTT, Modbus TCP, DNP3), HMI design, and industrial control systems. Quadrilingual (EN · ES · FR · IT). Dual Chilean/Bolivian citizenship.

EXPERIENCE

University of Kigali *Kigali, Rwanda* (Upcoming) 07/2026 – 08/2026

IoT- AI Based Soil and Environment Monitoring - Secondment

- ▶ Collaborating on a 1-hectare precision-agriculture pilot using five 7-in-1 soil sensors to monitor soil and environmental parameters.
- ▶ Designing wireless sensor-to-microcontroller data flow (TinyML on Arduino Nano 33 BLE Sense) with LoRa transmission to the cloud..
- ▶ Contributing to a lightweight ML model for predicting environmental conditions and crop yield/quality, visualised via a farmer-facing dashboard.

Siemens Connected Curriculum *Manchester, UK* 06/2026 – 07/2026
Summer Internship

- ▶ Selected signals for acquisition from the Festo CP Lab (mobile phone production line) and implemented OPC UA-based data acquisition.
- ▶ Built a Node-RED pipeline to stream data to Siemens Insights Hub, developing dashboards for oven energy consumption, per-order cycle time, data-quality errors, and operator-station cycle time by product type.
- ▶ Programmed the ABB YuMi dual-arm robot in RobotStudio using RAPID to develop and optimise motion sequences, assessing the feasibility of automating a candle-making process (wick placement, wax filling, and cup handling).

Grümm *Stockport, UK* 01/2026 – Present

Automation Engineering Intern

- ▶ Selected components and designed safety circuit ensuring compliance with safety standards.
- ▶ Programmed WAGO PLC (IEC 61131-3) for automated dispensing; designed HMI with PLC-to-HMI Ethernet comms.
- ▶ Integrated purchase order interface with PLC to automatically trigger device operations based on incoming orders.
- ▶ Conducted acceptance testing, debugged logic, and documented electrical design and maintenance protocols.
- ▶ Optimised code to improve cycle time and system responsiveness during testing.

Manchester Metropolitan University *Manchester, UK*
12/2025 – 05/2026

Research Assistant

- ▶ Programmed RTAC SEL 3555 PLC/gateway (SCD) for PID control over a simulated microgrid on OPAL RT.
- ▶ Mapped and addressed signals inside RTAC to enable proper data flow between control levels..
- ▶ Configured DNP3 bidirectional comms (OPAL RT ↔ RTAC) to build a two-level industrial control testbed.
- ▶ Programmed SCADA in Node-RED with bidirectional RTAC comms for real-time monitoring and signal mapping.

Alhona – Industry 4.0 Consulting *Bilbao, Spain* 09/2023 – 06/2025
IoT Project Manager

- ▶ Researched industrial protocols (MQTT, OPC UA, Modbus TCP, FTP) for optimal data acquisition per machine type.
- ▶ Defined signal architecture (selection, taxonomy, aliases) across 3 plants in Czech Republic and Poland.
- ▶ Led supplier negotiations and coordinated with international plant teams on-site to ensure alignment.

- ▶ Correlated machine parametrisation with cycle time variations.

- ▶ Found that average inter-order wait time was nearly double documented expectations, revealing a training-based improvement opportunity with direct business impact.

CEIT – Investigation Centre *San Sebastián, Spain* 09/2022 – 12/2022

Robotics Intern – Final Project

- ▶ Delivered and programmed in TMFlow 2 projects independently, earning a dissertation grade of 10/10 with Honours.
- ▶ Utilised TM5Omron robot vision system for tic-tac-toe with unbeatable logic; optimised game logic covering 20,000+ move combinations.
- ▶ Resolved interface issues and robot limitations, demonstrating rapid adaptation to proprietary programming environments.
- ▶ Developed portrait-sketching application.

EDUCATION

Manchester Metropolitan University *UK* 2025 – 2026

MSc Robotics & Automation

PLC prog., Virtual commissioning (TIA Portal, NX), Robotics (ROS, Python), AI, IIoT, Digital Manufacturing.

University of Navarra Tecnun *Spain* 2019 – 2023

BEng Industrial Electronic Engineering

4th best university in Spain · Times Higher Education

Politecnico di Milano *Italy* 02 – 07/2023

Exchange – Automation & Control Engineering

Top 20 in Europe for Engineering · QS Rankings

University of Navarra Tecnun *Spain* 2023 – 2024

Business Introduction Program

Project management, risk, leadership, strategic problem-solving.

SKILLS

PROGRAMMING LANGUAGES

Python · C/C++

· JavaScript · PASCAL · RAPID

PLC / CONTROL

TIA Portal · CODESYS · WAGO ·

AcSElerator · IEC 61131-3

PROTOCOLS

OPC UA · MQTT · Modbus TCP · DNP3

· S7 · Ethernet/IP

TOOLS & SOFTWARE

Node-RED · RT-LAB ·

RobotStudio · Siemens NX · Coppeliasim · EPLAN · Insights

Hub · KincoDTools (HMI design) · Fusion 360

TRANSVERSAL

Project Management · Data Analysis

· Signal Taxonomy · Hardware-in-the-Loop (HIL) · Technical

Documentation

AWARDS & COMPETITIONS

- ▶ WorldSkills UK – National Finalist, Industry 4.0 (2026)

- ▶ Siemens Xceletator Programme (2026)

LANGUAGES

ES native EN bilingual FR bilingual IT conversational

